



SAFETY Department
SIMHADRI

Note No :1



Ref:SMTTPP/SAFETY/2024-25/24/25-03/128468

Date:

14/05/2024(18:28:32)

Subject:Approval for LMI on Written Safe Work Procedures (LMI-OGN-OPS-SAFT/006)

Sirs,

As per requirement of clause 1.3 of LMI-OPS-SAFT-006 Rev0 - Written Safe Work Procedures, it is mandatory to review the LMI once in every 2 years in accordance with the respective i.e OGN COS-ISO-00-OGN/OPS-SAFT/006

Hence review has been done accordingly. However since there is no change in the respective OGN document, there is no change adopted in this LMI also.

Reviewed LMI is hereby put up for review and approval.

Enclosures:

1. OGN for written safe work procedures- OGN COS-ISO-00-OGN/OPS-SAFT/006
2. LMI for written safe work procedures- LMI-OPS-SAFT-006_Rev0

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Note No:2

reviewed and forwarded

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पुली सदाशिव रेड्डी - उप महाप्रबंधक

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Note No:3

May please be further reviewed for approval.

Submitted by:

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Note No:4

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Note No:5

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Note No:6

Reviewed and is being forwarded for further process.

Submitted by:

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Note No:7

It was instructed earlier to bring the LMI in main Pradip note and not as attachment.

May please see with regard to that.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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Note No:8

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Note No:9

Review and comments after discussion with in respective groups

Submitted by:

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Note No:10

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Note No:11

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Note No:12

Submitted by:

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Note No:13

Submitted by:

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Note No:16

Reviewed and forwarded for further approvals

Submitted by:

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Note No:18

Please review

Submitted by:

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Note No:19

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Submitted by:

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Note No:20

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Note No:21

Please review

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Note No:22

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Note No:23

Reviewed and forwarded for approval.

Submitted by:

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Note No:24

May please review LMI .

Submitted by:

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Note No:25

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Note No:26

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Note No:27

Please review the LMI

Submitted by:

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Note No:28

Please review

Submitted by:

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Note No:29

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Note No:30

Forwarded for approval.

Submitted by:

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Note No:31

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Submitted by:

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Note No:32

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Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

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Note No:33

May pl review

Submitted by:

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Note No:34

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Note No:35

Submitted by:

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Note No:36

Reviewed and forwarded.

Submitted by:

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Note No:37

Reviewed & forwarded.

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Note No:38

Please review.

Submitted by:

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Note No:39

Please review the LMI

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Note No:40

Forwarded for approval

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Note No:41

LMI is reviewed and forwarded for approval please.

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Note No:42

Submitted by:

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Note No:43

Submitted by:

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Note No:44

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Note No:45

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Note No:46

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Note No:47

Noted and forwarded for approval

Submitted by:

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Note No:48

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Note No:49

Submitted by:

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Note No:50

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

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On:24/07/2024(19:22:22) **Ref:**SMTPP/SAFETY/2024-25/24/25-03/128468



SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**Written Safe Work Procedures Of
Simhadri Super Thermal Power Station
LMI-OGN-OPS-SAFT/006 Rev: 00
May 2024**

Approved for Implementation by HOP (Simhadri)

	TITLE	Written Safe Work Procedures	SIMHADRI
	LMI NO	LMI/OGN/OPS/SAFT/006	

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	TITLE	Written Safe Work Procedures	SIMHADRI
	LMI NO	LMI/OGN/OPS/SAFT/006	

SAFETY DEPARTMENT

TITLE OF LMI:		Written safe work procedures of Simhadri Super Thermal Power Station					
LMI No:		LMI-OGN-OPS-SAFT/006					
ISSUE NO	REV NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
01	00	.03.22	All clauses are new as the document is being issued for the first time.	P S S REDDY	ALL HODs	GM(O&M)	HOP
02	00	.05.24	Document reviewed on 07.05.2024. No changes done.	M ARAVIND THANGAM	ALL HODs	GM(O&M)	HOP

	TITLE	Written Safe Work Procedures	SIMHADRI
	LMI NO	LMI/OGN/OPS/SAFT/006	

LMI APPROVAL DOCUMENT

<i>Title of LMI:</i>	Written Safe work Procedures of Simhadri Super Thermal Power Station	
<i>LMI No.:</i>	LMI-OGN-OPS-SAFT/006	
<i>Issue No. : 02</i>	<i>Revision: 0</i>	<i>Date : .05.2024</i>
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1.0 SUMMARY

1.1 Superseded documents	Not Applicable	
1.2 Scope	NTPC Simhadri Super Thermal Power Station.	
1.3 Review	Responsibility	Safety Department
	Review Frequency	2 yearly basis or as and when required, whichever is earlier
	Review Authority	Head of Project

2.0 RECORDS OF REVISION

NIL

3.0 INTRODUCTION

- a) All Written Safe Work Procedures shall be consistent with other types of established documentation, like Directives, Guidance Notes, Advisories, Rules. The purpose of such documentation is to ensure that all tasks with high-risk potential for adverse safety or occupational health related incidents are identified, analyzed and procedures compiled for the safe performance of tasks, to ensure that tasks are performed according to these procedures, and consequently assure a safe work environment.
- b) It is essential to note that the Directives, Guidance Notes or Information Notes are issued by the Corporate-OS from time to time to support consistency of effort in some aspects of the operational aspects of projects. Location Management Instructions (LMIs) are documents derived from such Directives, Guidance Notes or Information Notes, and are site-specific, mandatory documents. Corporate-OS cannot envisage every need of a location and issue guidance documentations for each such situation. Thus, LMIs are essential to ensure completeness of the documentation to cater to the needs of location specific requirements. Consequently, LMIs include Written Safe Work Procedures issued in different forms, as identified and required by the location management.

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- c) This LMI describes the system for written safe work procedures that is to be implemented in all the operational projects. The intent of written safe work procedures is to ensure a safe work environment for all persons working in the company's operating locations. The purpose of written safe work procedures is that all sites and business functions comply in a consistent manner with corporate policy, framework, directives and protocols

4.0 ROLES & ACCOUNTABILITY

- a) The HOP is accountable to ensure that, a system of formal Written Safe Work Procedures is implemented at the project, and a system is in place to assure its implementation in the form of periodic audits.
- b) In operational plants, Head (O&M) shall be accountable: to implement this guidance; to ensure safe work procedures are implemented; and, that all persons are trained in safe work procedures relevant to their work.
- c) For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for the above two roles.
- d) In construction projects, Head (Construction) shall be accountable: to ensure availability of adequate safe work procedures from contractors and to ensure that system is in place to verify and ensure contractors are complying to their respective written safe work procedures.
- e) Heads of Department (HOD)/Heads of Sections (HOS) shall be accountable for developing and maintaining a register(s) of written safe work procedures applicable for their section/department/function; shall be accountable for any non-compliance with such requirements; and, for ensuring persons in their respective department/function are trained in the relevant requirements.
- f) All employees are accountable to make themselves aware of and comply with their respective section/department/function's written safe work procedures relevant to their work.
- g) Persons responsible for setting off other people to work shall be accountable for ensuring that written safe work procedures relevant to that work, are implemented.

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5.0 SYSTEM OF WRITTEN SAFE WORK PROCEDURES

5.1 Types of documentation to ensure a safe work environment

5.1.1 Safety Rules for achieving Safety from the System

Achieving safety from the system derived hazards is the principal tool to ensure a safe work environment. Safety Rules governing methods of achieving control over hazards from the interconnected Electrical & Mechanical (E&M) system shall be clearly defined issued and made accessible to relevant employees wherever located.

Such E&M Safety Rules shall detail and chart the following requirements:

- Define the requirement for ensuring clear and unambiguous identification of plant and equipment.
- Clearly define the items of plant and equipment to which the rules apply and any limits in terms of boundary or termination points with instructions on what to do when isolations are needed across boundary points.
- Establish safety clearances for HV equipment and measures to ensure they are not infringed.
- Ensure that, where appropriate, measures for isolation; earthing; venting; draining; purging and release of stored energy have been taken before work or testing commences on plant and/or equipment.
- Provide guidance on securing points of isolation to prevent interference with measures to ensure that final control over the isolation remains with the worker(s).
- Specifying need to produce specialist procedures for essential live working or testing on electrical equipment or when work requires plant to be energized.
- Detail the requirement for formal nomination of trained and competent people to implement the safety rules and apply precautions; and also incorporate a means to ensure that they are aware of their responsibilities.
- Detail different types of safety documents under which work is executed: Permit to Work (PTW), Limited Work Certificate (LWC), etc.

Safety Rules meeting the above stipulations have been issued as Safety Rules (Electrical & Mechanical) Handbook. Further elaboration of the Implementation of the Safety Rules (Electrical & Mechanical) has been issued as an LMI. **– Responsibility: All depts.**

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5.1.2 Hazard Identification & Risk Assessments (HIRA):

Identification of Hazards is at the core of NTPC Safety Policy to ensure a safe work environment. Every location is required to assess 'risks' to the health and safety of anyone who may be affected by its activities (e.g. employees, contractors, visitors, members of the public). The basic requirement is to examine what could cause harm to people as a result of work activities and to make a judgment about whether enough precautions have been taken or, if further actions are required. It is mandatory to have in place written Hazard Identification & Risk Assessments.

Further elaboration of the implementation aspects of Hazard Identification and Risk Assessments has been issued as an LMI.

– Responsibility: All depts.

5.1.3 Standard Operating Procedures (SOPs)

- SOPs are documented procedures identified by the Location to ensure the effective planning and control of the operations processes. SOPs detail the practices, methods and steps to be adopted while carrying out routine & non-routine operations activities like High Voltage Switching.
- SOPs, by definition, form part of the Location Management Instructions (LMIs).
- The numbers of SOPs may vary depending on the location. For example, a large coal-fired station may have 100 SOPs while a small solar plant only 3 SOPs. While the numbers of SOPs at a location may vary as per that location's assessed needs, the requirement to have in place written SOPs is mandatory for every location.
- Identification of SOP Requirements: Each location must demonstrate evidence of a professional approach in identification of operating processes where relevant HOD/HOS lists out different operating processes systems, evaluates each against already available guidance in the form of Directives, Guidance notes or Information Notes.
- It is possible that a location may deem some operating processes are non-essential towards ensuring safe work environment and may consciously decide against having a formal SOP. However, the relevant HOD/HOS must be able to demonstrate the professional and systematic process of arriving at such decision. Annexure-1 provides illustration of a table to demonstrate the SOP identification process.

– Responsibility: All depts.

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5.1.4 Standard Maintenance Procedures (SMPs)

- SMPs are documented procedures identified by the Location to ensure the effective planning and control of the maintenance processes. SMPs detail the practices, methods and steps to be adopted while carrying out routine & non-routine maintenance activities like preventive, scheduled and breakdown maintenance works. SMPs thus are higher level maintenance procedures that cover policy and philosophy with respect to a class/type of equipment for example horizontal motors, power inverters or activities like overhauls, statutory boiler inspections.
- SMPs, by definition, form part of the Location Management Instructions (LMIs).
- The numbers of SMPs may vary depending on the location. For example, a large coal-fired station may have 100 SMPs while a small solar plant may have 1 SMP. While the numbers of SMPs at a location may vary as per that location's assessed needs, the requirement to have in place written SMPs is mandatory for every location.
- Identification of SMP Requirements: Each location must demonstrate evidence of a professional approach in identification of maintenance processes where the relevant HOD/HOS lists out different maintenance processes, evaluates each against already available guidance in the form of Directives, Guidance notes or Information Notes.
- It is possible that a location may deem some maintenance processes are non-essential towards ensuring safe work environment and may consciously decide against having a formal SMP. However, the relevant HOD/HOS must be able to demonstrate the professional and systematic process of arriving at such decision. Annexure-2 provides illustration of a table to demonstrate the SMP identification process. Annexure-3 provides an outline of a typical SMP.

– Responsibility: All Maintenance depts.

5.1.3 Specific Job Plans

- Specific Job Plans also called Work Instructions are derived from the SMPs. They list out jobs that are carried out primarily of repetitive nature including overhauls. For example, a job plan or a work instruction for a portable water pump would be derived from the location's SMP on Preventive Maintenance of Horizontal Motors.
- Persons attempting to work on a wrong equipment has been a cause of multiple accidents worldwide. Positive identification of equipment is one the most basic safety

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rules that is proven to prevent accidents. Each location should have specific job plans for each equipment. For example, Pump1 and Pump2 in a system that has two pumps must have two unique Job Plans. The content may be similar but uniquely identifiable. A specific job plan is generally expected to be uploaded in the central repository and included in work planning process or where needed can be attached as an electronic copy in a format like PDF that cannot be edited.

– Responsibility: All depts.

5.1.4 Specific Isolation Lists

- a) Safety Rules (Electrical & Mechanical) specify types and means of securing isolations for different circumstances. Specific Isolation Lists are derived from such specifications and requirements.
- b) Persons isolating wrong equipment has been a cause of many accidents worldwide. Positive identification of isolation points starts right from the preparation of specific isolation lists. These are isolation lists for works that are of repetitive nature, including overhauls. Similar to specific job plans, each location should have specific isolation lists for each equipment. For example, Pump1 and Pump2 in a system that has two pumps must have two unique isolation lists with isolation points mentioned like Pump1 Suction Valve rather than just Pump Suction Valve. A specific isolation is generally expected to be uploaded in the central repository and included in work planning process.

- Responsibility: Permit issuing dept.

5.1.5 Job Safety Analyses & Method Statements

- a) Job Safety Analysis (JSA) & Method Statements (MS) are similar in intent but vary in scope. JSA is a written procedure developed to identify the dangers of a particular task in order to reduce the risk of injury. A Method Statement is much more detailed and aims to provide adequate levels of information across all elements of the work in relation to health and safety having particular regard to any associated hazards / risks. Method Statement will consider 'what', 'when', 'where' and 'how' the work will be done. Foreseeable hazards should be identified, and risk assessments provided to

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demonstrate suitable control measures. The detail required must be proportionate to the level of risk.

- b) Further elaboration of the implementation aspects of Job Safety Analyses & Method Statements has been issued as a separate document.

Annexure-4 provides illustration of a table to track the status of Specific Job Plans, Specific Isolation Lists and Specific JSAs.

- Responsibility: All depts.

5.2 Awareness & Training

- a) HOD/ HOS shall ensure that employees in their function/department/section are aware of the requirements of their written safe work procedures. Training records shall be maintained, in a central repository and be accessible.
- b) In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person responsible for such a project shall be accountable for ensuring persons carrying out work in such locations are trained in relevant written safe work procedures and such records are maintained and accessible during any audit.

- Responsibility: All HODs/HOSs

6.0 AUDIT

Evaluation of the compliance with this Guidance shall be carried out at least once annually and can be part of a larger scope technical or other audits. The audit shall be a combination of verification of records, verification of implementation in the field and interviews.

a.**System:** Auditors shall validate the evidence of a system as specified under 5.0.

b.**Records:** Audit shall include verification of availability of the records detailed under 7.0

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c.The location should be able to demonstrate that it has initiated preparation of specific job plans and specific isolation lists. Training records of persons in written safe work procedures shall be verified at random from three different departments/functions and observations included in the audit report.

d.HOD/HOS shall be accountable for demonstrating compliance to the awareness of their respective section/department/function staff of their relevant safe work procedures, including training records.

e.**Interviews:** check with at least five employees carrying out roles that are responsible for ensuring safe work environment randomly selected from different functions on their awareness of safe work procedures relevant to their respective roles.

f.**Field Audit Checkpoints:** Auditors shall take a random sample of at least three activities from at least three different functions, and where applicable from different control rooms, to assess the implementation of the written safe work procedures.

g.Evidence of action initiated, including exemplary penal action, for failure to implement safe work procedures shall also be verified and included in the audit report.

h.All such audit/assessment records shall be maintained by the project-side audit coordinator, and uploaded as scanned electronic copies that cannot be edited like a secure PDF into a central repository.

i.In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person accountable for such a project shall be accountable for ensuring periodic audits of implementation of the philosophy and principles of this guidance are carried out. Records of such audits shall be maintained.

7.0 RECORDS TO BE MAINTAINED

- a) All HOD/HOS shall maintain, register of written safe work procedures relevant to their respective function/department/section; evidence of a system for checking implementation of the written safe work procedures; evidence of corrective action (including exemplary penal action) taken when safety control measures required as per were not implemented. Records of all self-assessment checks shall be uploaded in

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a central repository. Training records of persons in written safe work procedures relevant to their responsibility shall be maintained, in a central repository and be accessible.

- b) Any record that is maintained as an electronic copy shall be in a format like PDF, that cannot be edited.

ASSOCIATED DOCUMENTS

ANNEXURE 1 - SAMPLE SOP REGISTER ANNEXURE

ANNEXURE 2 - SAMPLE SMP REGISTER

ANNEXURE 3 - OUTLINE OF A TYPICAL STANDARD MAINTENANCE PROCEDURE (SMP)

ANNEXURE 4 - SAMPLE SPECIFIC JOB PLAN, ISOLATION LIST & JSA REGISTER

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ANNEXURE 1 - SAMPLE SOP REGISTER

NOTE: THIS IS A SAMPLE - GIVEN FOR ILLUSTRATIVE PURPOSE ONLY.
ACTUAL NUMBERING SYSTEM CAN BE AS PER LOCATION REQUIREMENTS.
EQUIPMENT ID TO BE CONSISTENT WITH ERP/SAP SYSTEM.

	FUNCTION: OPERATIONS			
#	OPERATING PROCESS	Reference Document	Document # & Name	Remarks
1	Main Plant Start-up	OEM Manuals	XXX/OPN/001 Stage-1 Unit-1 Start-up Procedure	
2		OEM Manuals	XXX/OPN/002 Stage-1 Unit-2 Start-up Procedure	
25	High Voltage Switching	[1] Safety Rules (E&M) Handbook. [2] OGN-OPS-SYST-007 Rev.01 Operational Switching	XXX/OPN/025 High Voltage Switching	
76	XXXX XXX xx	NA	NIL	Not required to make a specific document because...
	FUNCTION: HR & ADMIN			
1	Facilities Management	NA	XXX/HR/001 Facilities Management	

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ANNEXURE 2 - SAMPLE SMP REGISTER

NOTE: THIS IS A SAMPLE - GIVEN FOR ILLUSTRATIVE PURPOSE ONLY.

ACTUAL NUMBERING SYSTEM CAN BE AS PER LOCATION REQUIREMENTS. EQUIPMENT ID TO BE CONSISTENT WITH ERP/SAP SYSTEM.

#	MAINTENANCE PROCESS	Reference Document	Document # & Name	Remarks
	FUNCTION: MECHANICAL MAINTENANCE			
1	Steam Turbine Overhaul	NA	XXX/MMD/001 Stage-1 Unit-1 Steam Turbine Overhaul	
2		NA	XXX/MMD/002 Stage-1 Unit-2 Steam Turbine Overhaul	
76	XXXX XXX xx	NA	NIL	Not required to make a specific document because assessed risk is low.
	FUNCTION: ELECTRICAL MAINTENANCE			
10	Preventive Maintenance of Power Transformers	OGN/OPS/ELEC/019	XXX/EMD/015 PM of Power Transformers	
	FUNCTION: MEDICAL			
2	Maintenance Policy for Portable Medical Devices	NA	XXX/MED/002 Maintenance Policy for Portable Medical Devices	

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ANNEXURE-3 - OUTLINE OF A TYPICAL STANDARD MAINTENANCE PROCEDURE (SMP)

1.0 SCOPE & OBJECTIVES

This would cover the class/type of equipment that is covered under the SMP. Inclusions and exclusions, if any. For e.g., an SMP on 'Maintenance of Water bearing Civil Structures' may have exclusions like 'township water supply tanks' that are not located within the defined factory premises.

2.0 SUPERSEDED DOCUMENTS

Self-explanatory.

3.0 REVISION DETAILS

Self-explanatory.

4.0 ROLES & ACCOUNTABILITY

Accountable roles for system & execution ownership to be specified.

5.0 REQUIREMENTS

5.1 Legal Requirements: Identify applicable legal/statutory compliance requirements.

5.2 Safety, Health, Environment & Quality Requirements:

Identify quality requirements normally compliant with OEM advice, corporate standards, and industry standards.

Identify Safety, Health & Environment risks, and control measures for the same. Can cross-reference to HIRA and Aspect-Impact studies of the Section/Function.

5.3 Technical Requirements: Technical requirements for the SMP normally based on OEM manuals, warranty requirements (when applicable), the experience of the location, corporate guidance, etc. References should be listed as an Annexure for ease of audit.

5.4 Training & Awareness Requirements: will include recorded annual review with section staff and specify training requirements, if applicable.

6.0 RECORDS

Specify the records that need to be maintained for evidence of compliance. Specify if applicable, KPIs (Key Performance Indicators). Use the Safety Framework KPIs as a starting point.

7.0 COMPLIANCE AUDIT

Specify the audit guidelines & frequency.

8.0 REVIEW

Specify review frequency.

ANNEXURES - e.g., Reference documents like LMIs based on OD or OGN that are part of 5.3 Technical Requirements to be included for consolidated view, and other references when appropriate

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ANNEXURE 4 - SAMPLE SPECIFIC JOB PLAN, ISOLATION LIST & JSA REGISTER

NOTE: THIS IS A SAMPLE - GIVEN FOR ILLUSTRATIVE PURPOSE ONLY.

ACTUAL NUMBERING SYSTEM CAN BE AS PER LOCATION REQUIREMENTS. EQUIPMENT ID TO BE CONSISTENT WITH ERP/SAP SYSTEM.

#	Equipment Id	Equipment/Location ID	Job Plan Number	Specific JSA Number	Specific Isolation List Number	Remarks
	Function: MECHANICAL MAINTENANCE					
1	XxxxxxxxxX	Stage-1 Unit-1 Steam Turbine Overhaul	XXX/MMD/JP/00001	XXX/JSA/XXX/MMD//00001	XXX/OPN/WCD/00001	
2	XxxxxxxxxY	Stage-1 Unit-2 Steam Turbine Overhaul	XXX/MMD/JP/00002	XXX/JSA/XXX/MMD//00002	XXX/OPN/WCD/00002	
	Function: ELECTRICAL MAINTENANCE					
23	XxxxxxxxYx	Stage-3 Unit-1 Excitation Transformer-1	XXX/EMD/JP/00045	XXX/JSA/XXX/MMD//00045	XXX/OPN/WCD/00246	
	Function: MEDICAL					
1	ZxxxxxxxYx	Sterilization of hospital rooms	XXX/MED/JP/00001	XXX/JSA/XXX/MED//00012	NA	